



UBL HRC – API INTEGRATION MANUAL

VERSION: 3.1

Prepared by:
UNITED BANK LIMITED

CONTENTS

1. INTRODUCTION	4
1.1 OVERVIEW	4
1.2 DEFINITION OF TERMS	4
1.3 WEB SERVICE DESCRIPTION	4
2. SECURITY	5
MORE THAN JUST ENABLING COMMUNICATION BETWEEN UBL AND NON UBL SYSTEM, SECURITY IS ALSO A PRIMARY CONCERN FOR THE WEB SERVICE	5
THE FOLLOWING SECURITY MEASURES CAN BE MADE IN-EFFECT	5
3. SERVICES	6
3.1 ACCOUNT VALIDATION(Ws).....	6
3.1.1 Description	6
3.1.2 Service URL	6
Return Values.....	6
3.1.3 Request Parameters	7
3.1.4 Sample Request XML	7
3.1.5 Response Parameters (In case of UBL Account)	8
3.1.6 Sample Response (Successful)	8
3.1.7 Sample Response (Failure)	9
3.1.8 3. Response Fields (In case of other than UBL Account)	10
3.1.9 Sample Response (Successful)	10
3.1.10 Exception Response XML	11
3.2 BALANCE ENQUIRY(Ws).....	12
3.2.1 Description	12
3.2.2 Request Parameters	12
3.2.3 Sample Request XML	12
3.2.4 Response Parameters.....	13
3.2.5 Sample Response	13
3.3 POST TRANSACTION(Ws).....	14
3.3.1 Description	14
3.3.2 Request Parameters	15
3.3.3 Sample XML Request	17
3.3.4 Response Parameters.....	18
3.3.5 Sample Response	18
3.4 TRANSACTION STATUS (Ws).....	19
3.4.1 Description	19
3.4.2 Request Parameters	19
3.4.3 Sample Request XML	19
3.4.4 Response Parameters.....	20
3.4.5 Sample Response	20
4. APPENDICES.....	27
4.1 TRANSACTION STATUS CODE / RETURN MESSAGE.....	27
4.2 ERROR AND REJECTED STATUS CODE / RETURN MESSAGE	28
4.3 BANK CODE AND ACCOUNT FORMATS	30
4.4 AUTHENTICATION (OAUTH) STEPS	33

1. INTRODUCTION

1.1 Overview

UBL HRC API is an interface (XML Web Services) developed for UBL Remitting Agencies/Banks (Tie-ups) enabling them to send their remittances to UBL in a real time manner.

This document describes the technical details to integrate UBL Web Services with Remittance Booking systems running at the Tie-ups.

1.2 Definition of Terms

UBLWS

This is the acronym for UNITED BANK LIMITED Web Service.

Interface

This refers to a programmatic online interface. In this case, the UBL interface is a Web Service for tie-ups who want to process/send remittance transactions in an online and real time manner.

1.3 Web Service Description

The Web Service descriptor for Test can be accessed at

UAT URLS (Exposed over VPN and Internet):

<https://dpgwtdl.ubl.com.pk/tdl/sandbox/oauth/oauth2/token>
https://dpgwtdl.ubl.com.pk/tdl/sandbox/UBL_HRC_API_GetStatus
https://dpgwtdl.ubl.com.pk/tdl/sandbox/UBL_HRC_API_BalanceInquiry
https://dpgwtdl.ubl.com.pk/tdl/sandbox/UBL_HRC_API_FundTransfer
https://dpgwtdl.ubl.com.pk/tdl/sandbox/UBL_HRC_API_AccountValidation

Disclaimer:

Note that the above exposed URLs are subject to change.

Services Timeout value

The services times out in **180 seconds**.

2. SECURITY

More than just enabling communication between UBL and Non UBL System, Security is also a primary concern for the Web Service.

The following security measures can be made in-effect:

SSL

Secure Socket Layer Connection is a requirement. Invocation of the web service fails if transmission is done outside SSL.

Username/ Password / Authentication Code

Each tie-up will be assigned a username and a password to make calls to the UBLWS. Each call to the functions offered by the web service requires the username, password and Authentication code as parameters. This will ensure that although the web service is public, only the authenticated users can invoke it.

3. SERVICES

Account Validation^(WS)

This service allows user to validate any account maintained in UBL or any other bank account which is available with UBL for the purpose of IBFT.

3.1.1 Description

The service returns information associated with the given account. This service will be specifically useful for validating an account. The account may be in UBL or any other bank available on 1Link Network.

1. The user provides the Bank code, Branch code (optional) and Account number along with the UBLWS Security credentials.
2. UBLWS Validates the login info (User id, Password and Authentication Code)
3. UBLWS returns the success or failure status of the information submitted, on successful submission UBLWS will return the Account Detail.

3.1.2 Service URL

https://dpgwtdl.ubl.com.pk/tdl/sandbox/UBL_HRC_API_AccountValidation?wsdl

Return Values

- ❖ In Case of UBL Account
 - Branch Code
 - Branch Name
 - Branch City/Address
 - Title of Account
 - NIC Number
- ❖ In Case of Other Bank
 - Title of Account

* Please refer Appendix 4.2 for the list of bank codes.

3.1.3 Request Parameters

Tag	Type	Length	Mandatory For UBL/IBAN A/C	Mandatory For Other Bank	Description
<CustLoginId>	String	-	Yes	Yes	Customer Login Id
<CustPassword>	String	-	Yes	Yes	Customer Password
<Authen_Code>	String	-	Yes	Yes	Authentication Code
<BankCode>	integer		Yes	Yes	Bank Code
<BranchCode>	String	4	No	No	Branch Code
<AccountNumber>	String	UBL: 24 for IBAN 9 or 12 for regular Account Number Other Banks: see Section 4.3 Bank Code & Account Formats	Yes	Yes	Account Number or IBAN

3.1.4 Sample Request XML

```

<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ubl="http://localhost/UBL_HRC_API/"
xmlns:ubl1="http://UBL_HRC_API_AccountValidation_IFX_AccountValidation_In">
  <soapenv:Header/>
  <soapenv:Body>
    <ubl:Opr_AccountValidation>
      <ubl1:IFX>
        <SignonRq>
          <SignonPswd>
            <SignonRole></SignonRole>
            <CustId>
              <SPName></SPName>
              <CustPerId></CustPerId>
              <CustLoginId>100001</CustLoginId>
            </CustId>
            <CustPswd>
              <CryptType></CryptType>
              <SecObjId></SecObjId>
              <CryptPswd>
                <BinLength>0</BinLength>
                <CustPassword> xxxxxxxx </CustPassword>
                <ubl1:Authen_code> xxxxxxxx </ubl1:Authen_code>
              </CryptPswd>
            </CustPswd>
            <GenSessKey></GenSessKey>
          </SignonPswd>
          <SessKey></SessKey>
          <ClientDt></ClientDt>
          <CustLangPref></CustLangPref>
          <ClientApp>
            <Org></Org>
            <ServiceID></ServiceID>
            <Name>AccountValidation</Name>
            <Version></Version>
          </ClientApp>
        </SignonRq>
      </ubl1:IFX>
    </ubl:Opr_AccountValidation>
  </soapenv:Body>
</soapenv:Envelope>

```

```

    <ubl1:IP></ubl1:IP>
    <ubl1:DeviceId></ubl1:DeviceId>
    <ubl1:AppId></ubl1:AppId>
  </ClientApp>
  <SuppressEcho></SuppressEcho>
</SignonRq>
<BankSvcRq>
  <RqUID></RqUID>
  <SvcName></SvcName>
  <ubl1:AccountValidationRq>
    <ubl1:RqUID></ubl1:RqUID>
    <BankCode>24</BankCode>
    <BranchCode>0605</BranchCode>
    <ubl1:AccountNumber>10947707</ubl1:AccountNumber>
  </ubl1:AccountValidationRq>
</BankSvcRq>
</ubl1:IFX>
</ubl:Opr_AccountValidation>
</soapenv:Body>
</soapenv:Envelope>

```

3.1.5 Response Parameters (In case of UBL Account)

Tag	Type	Description
<StatusCode>	String	Status Code
<StatusDesc>	String	Status Description
<BranchCode>	String	Branch Code
<BranchName>	String	Branch Name
<City>	String	Branch Address
<CNIC>	String	NIC Number
<TitleOfAccount>	String	Title of Account

3.1.6 Sample Response (Successful)

```

<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <Opr_AccountValidationResponse xmlns="http://localhost/UBL_HRC_API/">
      <part>
        <ns0:IFX
          xmlns:ns0="http://UBL_HRC_API_AccountValidation_IFX_AccountValidation_Out">
          <SignonRs>
            <ClientDt>4/18/2012 1:56:07 PM</ClientDt>
          </SignonRs>
          <BankSvcRs>
            <AccountValidationRs>
              <Status>
                <StatusCode>100</StatusCode>
                <StatusDesc>Success</StatusDesc>
              </Status>
              <RqUID>1bcc38a4-49dd-42c1-b919-fc012c1f1b43</RqUID>
            </Response>
          </AccountValidationRs>
        </BankSvcRs>
      </part>
    </Opr_AccountValidationResponse>
  </soap:Body>
</soap:Envelope>

```

```

<BranchCode>605</BranchCode>
<BranchName>CITY BRANCH, KARACHI</BranchName>
<City>UBL BUILDING, I.I.CHUNDRIGAR ROAD, PO BOX NO 5221,</City>
<CNIC></CNIC>
<TitleOfAccount>SUHAIL AHMED</TitleOfAccount>
</Response>
</AccountValidationRs>
</BankSvcRs>
</ns0:IFX>
</part>
</Opr_AccountValidationResponse>
</soap:Body>
</soap:Envelope>

```

3.1.7 Sample Response (Failure)

```

<?xml version="1.0" encoding="utf-8"?>
<soap: Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <Opr_AccountValidationResponse xmlns="http://localhost/UBL_HRC_API/">
      <part>
        <ns0: IFX
xmlns:ns0="http://UBL_HRC_API_AccountValidation_IFX_AccountValidation_Out">
          <SignonRs>
            <ClientDt>4/18/2012 2:13:05 PM</ClientDt>
          </SignonRs>
          <BankSvcRs>
            <AccountValidationRs>
              <Status>
                <StatusCode>1017</StatusCode>
                <StatusDesc>Invalid Account Number or Branch Code.</StatusDesc>
              </Status>
              <RqUID>0fe1975f-5e88-4db8-a068-7e0ecbb8938d</RqUID>
              <Response>
                <BranchCode></BranchCode>
                <BranchName></BranchName>
                <City></City>
                <CNIC></CNIC>
                <TitleOfAccount></TitleOfAccount>
              </Response>
            </AccountValidationRs>
          </BankSvcRs>
        </ns0: IFX>
      </part>
    </Opr_AccountValidationResponse>
  </soap:Body>
</soap:Envelope>

```

3.1.8 Response Fields (In case of other than UBL Account)

Tag	Type	Description
<StatusCode>	String	Status Code
<StatusDesc>	String	Status Description
<TitleOfAccount>	String	Title of Account

3.1.9 Sample Response (Successful)

```
<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <Opr_AccountValidationResponse xmlns="http://localhost/UBL_HRC_API/">
      <part>
        <ns0:IFX
  xmlns:ns0="http://UBL_HRC_API_AccountValidation_IFX_AccountValidation_Out">
          <SignonRs>
            <ClientDt>4/18/2012 2:18:04 PM</ClientDt>
          </SignonRs>
          <BankSvcRs>
            <AccountValidationRs>
              <Status>
                <StatusCode>100</StatusCode>
                <StatusDesc>Success</StatusDesc>
              </Status>
              <RqUID>70a9feca-411f-4d65-baaa-c5dc88231c6f</RqUID>
              <Response>
                <TitleOfAccount>Simulator One Link</TitleOfAccount>
              </Response>
            </AccountValidationRs>
          </BankSvcRs>
        </ns0:IFX>
      </part>
    </Opr_AccountValidationResponse>
  </soap:Body>
</soap:Envelope>
```

3.1.10 Exception Response XML

In the case of Authentication Failed, the service provides status code and description in the Status of **SignonRs** Fields.

```
<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <Opr_AccountValidationResponse xmlns="http://localhost/UBL_HRC_API/">
      <part>
        <ns0:IFX
xmlns:ns0="http://UBL_HRC_API_AccountValidation_IFX_AccountValidation_Out">
          <SignonRs>
            <ClientDt>4/18/2012 2:22:12 PM</ClientDt>
          </SignonRs>
          <BankSvcRs>
            <Status>
              <StatusCode>1020</StatusCode>
              <StatusDesc>Provided Information is Incorrect.</StatusDesc>
            </Status>
            <RqUID>0c36d6c9-bc09-4bce-b2f1-6ebf61623111</RqUID>
          </BankSvcRs>
        </ns0:IFX>
      </part>
    </Opr_AccountValidationResponse>
  </soap:Body>
</soap:Envelope>
```

Balance Enquiry(WS)

This service returns Account balance (in figures) for any specific account type.

3.2.1 Description

User provides the account type (NR account or Main Disbursement Account) to the service and in return, the service returns actual account balance.

Service URL

https://dpgwtdl.ubl.com.pk/tld/sandbox/UBL_HRC_API_BalanceInquiry?wsdl

3.2.2 Request Parameters

Tag	Type	Length	Mandatory	Description
<CustLoginId>	String	-	Yes	Customer Login Id
<CustPassword>	String	-	Yes	Customer Password
<Authen_Code>	String	-	Yes	Authentication Code
<AccountType>	Integer	1	Yes	1= MDA Account / 2 =NR Account

3.2.3 Sample Request XML

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ubl="http://localhost/UBL_HRC_API/"
xmlns:ubl1="http://UBL_HRC_API_BalanceInquiry_IFX_BalanceInquiry_In">
  <soapenv:Header/>
  <soapenv:Body>
    <ubl:Opr_BalanceInquiry>
      <ubl1:IFX>
        <SignonRq>
          <SignonPswd>
            <SignonRole></SignonRole>
            <CustId>
              <SPName></SPName>
              <CustPermId></CustPermId>
              <CustLoginId>100001</CustLoginId>
            </CustId>
            <CustPswd>
              <CryptType></CryptType>
              <SecObjId></SecObjId>
              <CryptPswd>
                <BinLength>0</BinLength>
                <CustPassword>xxxxxxx</CustPassword>
                <ubl1:Authen_code>xxxxxxx</ubl1:Authen_code>
              </CryptPswd>
            </CustPswd>
            <GenSessKey></GenSessKey>
          </SignonPswd>
          <SessKey></SessKey>
          <ClientDt></ClientDt>
          <CustLangPref></CustLangPref>
          <ClientApp>
            <Org></Org>
            <ServiceID></ServiceID>
            <Name></Name>
            <Version></Version>
            <ubl1:IP></ubl1:IP>
            <ubl1:DeviceId></ubl1:DeviceId>
          </ClientApp>
        </SignonRq>
      </ubl1:IFX>
    </ubl:Opr_BalanceInquiry>
  </soapenv:Body>
</soapenv:Envelope>
```

```

        <ubl1:Appld></ubl1:Appld>
      </ClientApp>
    <SuppressEcho></SuppressEcho>
  </SignonRq>
  <BankSvcRq>
    <RqUID></RqUID>
    <SvcName></SvcName>
    <ubl1:BalanceInquiryRq>
      <ubl1:RqUID></ubl1:RqUID>
      <AccountType>1</AccountType>
    </ubl1:BalanceInquiryRq>
  </BankSvcRq>
</ubl1:IFX>
</ubl:Opr_BalanceInquiry>
</soapenv:Body>
</soapenv:Envelope>

```

3.2.4 Response Parameters

Tag	Type	Description
<StatusCode>	String	Status Code
<StatusDesc>	String	Status Description
<FundsAvailable>	NUMBER	Account Balance (Amount)

3.2.5 Sample Response

```

<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <Opr_BalanceInquiryResponse xmlns="http://localhost/UBL_HRC_API/">
      <part>
        <ns0: IFX xmlns:ns0="http://UBL_HRC_API_BalanceInquiry_IFX_BalanceInquiry_Out">
          <SignonRs>
            <ClientDt>4/18/2012 2:29:16 PM</ClientDt>
          </SignonRs>
          <BankSvcRs>
            <BalanceInquiryRs>
              <Status>
                <StatusCode>100</StatusCode>
                <StatusDesc>Success</StatusDesc>
              </Status>
              <RqUID>8e88936b-dfa2-413f-a029-b8efbe6212b8</RqUID>
              <Response>
                <FundsAvailable>68,023,411.25</FundsAvailable>
              </Response>
            </BalanceInquiryRs>
          </BankSvcRs>
        </ns0: IFX>
      </part>
    </Opr_BalanceInquiryResponse>
  </soap:Body>
</soap:Envelope>

```

Post Transaction(WS)

Using this service, user can post transactions of any product type to CMDS directly without any delay. Following product types are included

Product Type	Description	Tran. Mode
D	Direct Deposit (UBL to UBL)	Online
A	B2C Transactions	Online
C- (IBFT enable)	Direct Deposit (Other banks)	Online
C	PRISM/Tez Raftaar Cheque	Offline
O	Prism/Tez Raftaar Cash	Offline

Note that the A type transactions are treated as D type, so all A type transactions should be sent with Bank Code 24.

3.3.1 Description

This service allows user to post transactions of various product types (listed above) to CMDS. Transactions are performed in online and offline modes.

Offline mode transactions are always queued and are batched for processing later on. In case of discrepancy in data, these transactions can be repaired and reprocessed by UBL.

In Online mode, system will post the transactions in real time manner and final status of these transactions (successful/rejected) will be returned to user. In case of any discrepancy in data the option of **Queued Up** or **Upfront Rejection** is available. API Consumer has to preemptively choose Upfront Rejection OR Queued Up mechanism for the processing of their online mode transactions.

If the Upfront Rejection mode is used the transactions are not queued and the consumer can itself retry after rectifying data issues. On the other hand if the Queued Up mode is used, the concerned transactions are parked at UBL end and are reposted after necessary repairing of data by UBL.

<ACCOUNTNO> parameter below in the Fund Transfer request represents; Beneficiary Account No. or IBAN.

For D type transactions, that are UBL to UBL, valid account number can be:

- 9 or 12 digits numeric string
OR
- 24 characters long alpha-numeric IBAN

For C type transactions (IBFT Enabled), that are UBL to Other Banks, valid account number can be:

- According to the format – see [Section 4.3 Bank Code & Account Formats](#)
OR
- 24 characters long alpha-numeric IBAN

NOTE: Account Numbers can be validated separately through Account Validation service mentioned overhead.

Service URL

https://dpgwtdl.ubl.com.pk /tdl/sandbox/UBL_HRC_API_FundTransfer?wsdl

3.3.2 Request Parameters

Tag	Data Type	Length	Description	Mandatory D Type	Mandatory C Type	Mandatory O Type	Mandatory A Type
<TrnRefNo>	String	16	Unique Transaction ID	Yes	Yes	Yes	Yes
<REMITTANCETYPE>	String	3	Product Type	Yes/ Value is (D)	Yes/ Value is (C)	Yes/ Value is (O)	Yes/ Value is (A)
<ISIBFT>	integer	1	1 in case of IBFT transaction and 0 for none IBFT	Value =0	Value =0/1	Value =0	Value =0/1
<SENDERNAME>	String	100	Sender name	Yes	Yes	Yes	Yes
<SENDERADDRESS1>	String	30	Sender address	Optional	Optional	Optional	Optional
<SENDERADDRESS2>	String	30	Sender address	Optional	Optional	Optional	Optional
<SENDERPOSTALCODE>	String	10	Sender Postal code	Optional	Optional	Optional	Optional
<SENDERCITY>	String	30	Sender city	Optional	Optional	Optional	Optional
<SENDERCOUNTRY>	String	30	Sender country	Optional	Optional	Optional	Optional
<SENDERPHONENO>	String	30	Sender phone no	Optional	Optional	Optional	Optional
<SENDEREMAILID>	String	30	Sender ID/EMAIL ID	Optional	Optional	Optional	Optional
<SENDERMESSAGE>	String	60	Comments	Optional	Optional	Optional	Optional
<REMITTANCEDATE>	Date	Format: yyyy-MM-ddTHH:mm:ss	Remittance booking date. Sample: 2017-01-24T18:55:50	Yes	Yes	Yes	Yes
<EXCHANGERATE>	Number		Exchange rate	Yes	Yes	Yes	Yes
<REMITTANCECURRENCYCODE>	String	3	Sender currency Codes	Optional	Optional	Optional	Optional
<REMITTANCEAMOUNT>	Number		Booking amount	Yes	Yes	Yes	Yes
<LOCALAMOUNT>	Number		Beneficiary Amount	Yes	Yes	Yes	Yes
<BENEFICIARYNAME>	String	100	Beneficiary name	Yes	Yes	Yes	Yes
<BENEFICIARYNICNO>	String	30	Beneficiary CINC Number	Optional	Optional	Optional	Optional
<ACCOUNTNO>	String		Beneficiary Account No. or IBAN – details in description above	Yes	Yes(for IBFT)	Optional	Yes(for IBFT)
<BANKCODE>	Number		Beneficiary bank Code	Yes	Yes	Yes	Yes (for IBFT)
<BANKNAME>	String	30	Beneficiary bank Name	Optional	Optional	Optional	Optional
<BRANCHCODE>	String	4	Beneficiary branch code	Optional	Optional	Optional	Optional
<BRANCHNAME>	String	90	Beneficiary branch name	Optional	Optional	Optional	Optional
<BRANCHCITY>	String	30	Beneficiary branch city	Optional	Optional	Optional	Optional
<BENEFICIARYADDRESS1>	String	100	Beneficiary Address	Optional	Optional	Optional	Optional

<BENEFICIARYADDRESS2>	String	100	Beneficiary Address	Optional	Optional	Optional	Optional
<BENEFICIARYADDRESS3>	String	100	Beneficiary Address	Optional	Optional	Optional	Optional
<BENEFICIARYPOSTALCODE>	String	10	Beneficiary postal Code	Optional	Optional	Optional	Optional
<BENEFICIARYCITY>	String	30	Beneficiary city	Optional	Optional	Optional	Optional
<BENEFICIARYCOUNTRY>	String	30	Beneficiary country	Optional	Optional	Optional	Optional
<BENEFICIARYPHONENO>	String	30	Beneficiary phone No	Optional	Optional	Optional	Optional
<BENEFICIARYMOBILENO>	String	30	Beneficiary mobile no	Optional	Optional	Optional	Optional
<BENEFICIARYEMAILID>	String	30	Beneficiary email address	Optional	Optional	Optional	Optional
<REMARKS>	String	100	Remarks/ comments	Optional	Optional	Optional	Optional
<ACCOUNTTITLE>	String	60	Account title	Optional	Optional	Optional	Optional
<USER_ID>	String	30	User Id provided to service user	Optional	Optional	Optional	Optional
<AUTHENTICATION_AGENT_CODE>	String	30	Authentication code provided to user	Optional	Optional	Optional	Optional
<Membership_Entity>	String	100	AML Field	Optional	Optional	Optional	Optional
<Member_Registration>	String	100	AML Field	Optional	Optional	Optional	Optional
<Remitter_DOB>	String	Format: dd/mm/yyyy	AML Field	Optional	Optional	Optional	Optional
<Remitter_NICN>	String	100	AML Field	Optional	Optional	Optional	Optional
<Remitter_Nationality>	String	100	AML Field	Optional	Optional	Optional	Optional
<Membership_Number>	String	100	AML Field	Optional	Optional	Optional	Optional
<Remitter_Father_Name>	String	100	AML Field	Optional	Optional	Optional	Optional
<AdditionalParam1>	String			Optional	Optional	Optional	Optional
<AdditionalParam2>	String	100	Relationship with Beneficiary	Optional	Optional	Optional	Optional
<PurposeOfPayment>	String			Optional	Optional	Optional	Optional
<PaymentCode>	String			Optional	Optional	Optional	Optional
<B2C_CNIC>	String	30		Optional	Optional	Optional	Yes
<B2C_TRANTYPE_ID>	Number			Optional	Optional	Optional	Yes

3.3.3 Sample XML Request

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ubl="http://localhost/UBL_HRC_API/"
xmlns:ubl1="http://UBL_HRC_API_FundTransfer_IFX_FundTransfer_In">
  <soapenv:Header/>
  <soapenv:Body>
    <ubl:Opr_FundTransfer>
      <ubl1:IFX>
        <SignonRq>
          <SignonPswd>
            <SignonRole></SignonRole>
            <CustId>
              <SPName></SPName>
              <CustPermId></CustPermId>
              <CustLoginId>100001</CustLoginId>
            </CustId>
            <CustPswd>
              <CryptType></CryptType>
              <SecObjId></SecObjId>
              <CryptPswd>
                <BinLength>0</BinLength>
                <CustPassword>xxxxxxxx</CustPassword>
                <ubl1:Authen_code>xxxxxxxx</ubl1:Authen_code>
              </CryptPswd>
            </CustPswd>
            <GenSessKey></GenSessKey>
          </SignonPswd>
          <SessKey></SessKey>
          <ClientDt></ClientDt>
          <CustLangPref></CustLangPref>
          <ClientApp>
            <Org></Org>
            <ServiceID></ServiceID>
            <Name></Name>
            <Version></Version>
            <ubl1:IP></ubl1:IP>
            <ubl1:DeviceId></ubl1:DeviceId>
            <ubl1:AppId></ubl1:AppId>
          </ClientApp>
          <SuppressEcho></SuppressEcho>
        </SignonRq>
        <BankSvcRq>
          <RqUID></RqUID>
          <SvcName></SvcName>
          <ubl1:FundTransferRq>
            <ubl1:RqUID></ubl1:RqUID>
            <TrnRefNo>SFTN038998</TrnRefNo>
            <REMITTANCETYPE>C</REMITTANCETYPE>
            <ubl1:ISIBFT></ubl1:ISIBFT>
            <ubl1:SENDERNAME>SAIMA AHSAN</ubl1:SENDERNAME>
            <ubl1:SENDERADDRESS1>4716 Windrush Cir</ubl1:SENDERADDRESS1>
            <ubl1:SENDERADDRESS2></ubl1:SENDERADDRESS2>
          </ubl1:FundTransferRq>
        </BankSvcRq>
      </ubl1:IFX>
    </ubl:Opr_FundTransfer>
  </soapenv:Body>
</soapenv:Envelope>
```

```

<ubl1:SENDERPOSTALCODE>73072</ubl1:SENDERPOSTALCODE>
<ubl1:SENDERCITY>Norman</ubl1:SENDERCITY>
<ubl1:SENDERCOUNTRY>usa</ubl1:SENDERCOUNTRY>
<ubl1:SENDERPHONENO></ubl1:SENDERPHONENO>
<ubl1:SENDEREMAILID></ubl1:SENDEREMAILID>
<ubl1:SENDERMESSAGE></ubl1:SENDERMESSAGE>
<ubl1:REMITTANCEDATE>2012-03-14T00:00:00</ubl1:REMITTANCEDATE>
<ubl1:EXCHANGERATE></ubl1:EXCHANGERATE>
<ubl1:REMITTANCECURRENCYCODE>GBP</ubl1:REMITTANCECURRENCYCODE>
<ubl1:REMITTANCEAMOUNT>1.00</ubl1:REMITTANCEAMOUNT>
<ubl1:LOCALAMOUNT>92.50</ubl1:LOCALAMOUNT>
<ubl1:BENEFICIARYNAME>KASHIF ANWER</ubl1:BENEFICIARYNAME>
<ubl1:BENEFICIARYNICNO>3520224311571</ubl1:BENEFICIARYNICNO>
<ubl1:ACCOUNTNO>0081001148016</ubl1:ACCOUNTNO>
<ubl1:BANKCODE>4</ubl1:BANKCODE>
<ubl1:BANKNAME></ubl1:BANKNAME>
<ubl1:BRANCHCODE>1085</ubl1:BRANCHCODE>
<ubl1:BRANCHNAME>BANK ALHABIB LIMITED ,MAIN BRANCH,MALL ROAD,LAHORE</ubl1:BRANCHNAME>
<ubl1:BRANCHCITY></ubl1:BRANCHCITY>
<ubl1:BENEFICIARYADDRESS1>HOUSE NO 1079/A FASEEL ROAD INSIDE</ubl1:BENEFICIARYADDRESS1>
<ubl1:BENEFICIARYADDRESS2></ubl1:BENEFICIARYADDRESS2>
<ubl1:BENEFICIARYADDRESS3></ubl1:BENEFICIARYADDRESS3>
<ubl1:BENEFICIARYPOSTALCODE></ubl1:BENEFICIARYPOSTALCODE>
<ubl1:BENEFICIARYCITY>LAHORE</ubl1:BENEFICIARYCITY>
<ubl1:BENEFICIARYCOUNTRY>PAK</ubl1:BENEFICIARYCOUNTRY>
<ubl1:BENEFICIARYPHONENO>5174975</ubl1:BENEFICIARYPHONENO>
<ubl1:BENEFICIARYMOBILENO></ubl1:BENEFICIARYMOBILENO>
<ubl1:BENEFICIARYEMAILID>fareed77@hotmail.com</ubl1:BENEFICIARYEMAILID>
<ubl1:REMARKS></ubl1:REMARKS>
<ubl1:ACCOUNTTITLE></ubl1:ACCOUNTTITLE>
<ubl1:USER_ID></ubl1:USER_ID>
<ubl1:AUTH_AGENT_CODE></ubl1:AUTH_AGENT_CODE>
<ubl1:Membership_Entity></ubl1:Membership_Entity>
<ubl1:Member_Registration></ubl1:Member_Registration>
<ubl1:Rmitter_DOB></ubl1:Rmitter_DOB>
<ubl1:Rmitter_NICN></ubl1:Rmitter_NICN>
<ubl1:Rmitter_Nationality></ubl1:Rmitter_Nationality>
<ubl1:Membership_Number></ubl1:Membership_Number>
<ubl1:Rmitter_Father_Name></ubl1:Rmitter_Father_Name>
<ubl1:AdditionalParam1></ubl1:AdditionalParam1>
<ubl1:AdditionalParam2></ubl1:AdditionalParam2>
<ubl1:PurposeOfPayment></ubl1:PurposeOfPayment>
<ubl1:PaymentCode></ubl1:PaymentCode>
<ubl1:B2C_CNIC></ubl1:B2C_CNIC>
<ubl1:B2C_TRANTYPE_ID></ubl1:B2C_TRANTYPE_ID>
</ubl1:FundTransferRq>
</BankSvcRq>
</ubl1:IFX>
</ubl:Opr_FundTransfer>
</soapenv:Body>
</soapenv:Envelope>

```

3.3.4 Response Parameters

Tag	Type	Description
<StatusCode>	String	Status Code
<StatusDesc>	String	Status Description

3.3.5 Sample Response

```

<?xml version="1.0" encoding="utf-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <Opr_FundTransferResponse xmlns="http://localhost/UBL_HRC_API/">
      <part>
        <ns0:IFX xmlns:ns0="http://UBL_HRC_API_FundTransfer_IFX_FundTransfer_Out">
          <SignonRs>
            <ClientDt>4/18/2012 2:52:57 PM</ClientDt>
          </SignonRs>
          <BankSvcRs>
            <FundTransferRs>
              <Status>
                <StatusCode>100</StatusCode>
                <StatusDesc>Success</StatusDesc>
              </Status>
              <RqUID>82da4b3d-7be9-42fd-be79-fa931199ff1f</RqUID>
            </FundTransferRs>
          </BankSvcRs>
        </ns0:IFX>
      </part>
    </Opr_FundTransferResponse>
  </soap:Body>
</soap:Envelope>

```

Transaction Status (WS)

This service allows users to get the current status of any transaction.

3.4.1 Description

The service returns Status code and Status Description of any transaction. User provides the Transaction Reference Number to the service.

Service URL

https://dpgwtdl.ubl.com.pk /tdl/sandbox/UBL_HRC_API_GetStatus?wsdl

3.4.2 Request Parameters

Tag	Type	Length	Mandatory	Description
<CustLoginId>	String	-	Yes	Customer Login Id
<CustPassword>	String	-	Yes	Customer Password
<Authen_Code>	String	-	Yes	Authentication Code
<REMITTERUNIQUEID>	Number	16	Yes	Transaction Reference Number
<AdditionalParam1>	String	-	No	
<AdditionalParam2>	String	-	No	

3.4.3 Sample Request XML

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:ubl="http://localhost/UBL_HRC_API/"
  xmlns:ubl1="http://UBL_HRC_API_GetStatus_IFX_GetStatus_In">
  <soapenv:Header/>
  <soapenv:Body>
    <ubl:Opr_GetStatus>
      <ubl1:IFX>
        <SignonRq>
          <SignonPswd>
            <SignonRole></SignonRole>
            <CustId>
              <SPName></SPName>
              <CustPerId></CustPerId>
              <CustLoginId>100001</CustLoginId>
            </CustId>
            <CustPswd>
              <CryptType></CryptType>
              <SecObjId></SecObjId>
              <CryptPswd>
                <BinLength>0</BinLength>
                <CustPassword>xxxxxxx</CustPassword>
                <ubl1:Authen_code>xxxxxxx</ubl1:Authen_code>
              </CryptPswd>
            </CustPswd>
            <GenSessKey></GenSessKey>
          </SignonPswd>
          <SessKey></SessKey>
          <ClientDt></ClientDt>
          <CustLangPref></CustLangPref>
          <ClientApp>
            <Org></Org>
            <ServiceID></ServiceID>
            <Name></Name>
            <Version></Version>
            <ubl1:IP></ubl1:IP>
          </ClientApp>
        </SignonRq>
      </ubl1:IFX>
    </ubl:Opr_GetStatus>
  </soapenv:Body>
</soapenv:Envelope>
```

```

        <ubl1:DeviceId></ubl1:DeviceId>
        <ubl1:Appld></ubl1:Appld>
    </ClientApp>
    <SuppressEcho></SuppressEcho>
</SignonRq>
<BankSvcRq>
    <RqUID></RqUID>
    <SvcName></SvcName>
    <ubl1:GetStatusRq>
        <ubl1:RqUID></ubl1:RqUID>
        <REMITTERUNIQUEID>SFTN038998</REMITTERUNIQUEID>
        <ubl1:AdditionalParam1></ubl1:AdditionalParam1>
        <ubl1:AdditionalParam2></ubl1:AdditionalParam2>
    </ubl1:GetStatusRq>
</BankSvcRq>
</ubl1:IFX>
</ubl:Opr_GetStatus>
</soapenv:Body>
</soapenv:Envelope>

```

3.4.4 Response Parameters

Tag	Type	Description
<StatusCode>	String	Status Code
<StatusDesc>	String	Status Description

3.4.5 Sample Response

```

<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <Opr_GetStatusResponse xmlns="http://localhost/UBL_HRC_API/">
      <part>
        <ns0:IFX xmlns:ns0="http://UBL_HRC_API_GetStatus_IFX_GetStatus_Out">
          <SignonRs>
            <ClientDt>4/18/2012 3:11:55 PM</ClientDt>
          </SignonRs>
          <BankSvcRs>
            <GetStatusRs>
              <RqUID>99989a8a-2037-4344-94f8-0a65866514c3</RqUID>
              <Response>
                <StatusCode>9</StatusCode>
                <StatusDesc>Processing</StatusDesc>
              </Response>
            </GetStatusRs>
          </BankSvcRs>
        </ns0:IFX>
      </part>
    </Opr_GetStatusResponse>
  </soap:Body>

```

4. APPENDICES

4.1 Transaction Status Code / Return Message

STATUS CODE	DESCRIPTION
100	Success - (Message received and processed successfully)
1034	Transaction Queued Up at UBL. - (Transaction Received at UBL)
1060	B2C TRANSACTION - (B2C Transaction Received at UBL)

Following transaction product wise status codes will be available for enquiry service once they are received at UBL end (100 - Success/ 1034-Transaction queued up at UBL)

STATUS Code	IBFT,PRI & TRC - C Type	COC - O Type	Direct Deposit - D Type
1	Pending	Pending	Pending
2	Pending	Pending	Pending
3	Pending	Pending	Pending
4	Pending	Pending	Pending
5	Pending	Pending	Pending
6	Processing	Processing	Processing
7	Rejected	Rejected	Rejected
8	Pending	Pending	Pending
9	Successful IBFT & TRC (Final Status)	Final Status	Processing
12	-	Final Status	Account Credited (Final Status)
13	-	-	-
14	-	-	-
15	-	Cash Picked Up (Final Status)	-
23	Staged Transaction Delete - Approved by Checker (Final Status)	Staged Transaction Delete - Approved by Checker (Final Status)	Staged Transaction Delete - Approved by Checker (Final Status)
109	-	Available for Payment	-
110	-	Processing	-
111	-	Processing	-
112	-	Processing	-
115	-	Pending	-
116	-	Pending	-
117	-	Pending	-
121	-	Pending	-
122	-	Pending	-
123	-	Pending	-

124	-	Pending	-
125	-	Cancelled (final status)	-
127	-	Pending	Pending
128	-	Cash transaction Converted to Pardes Card (Final Status)	-
131	-	COC - Rejected by Compliance (Final Status)	-
301	Pending	-	-
302	Pending	-	-
303	Pending	-	-
304	Pending	-	-
305	Processing	-	-
306	Pending	-	-
307	Pending	-	-
400	Pending	-	-
401	Pending	-	-
403	RTGS (Final Status)	-	-
900	Pending	Pending	-
901	Pending	Pending	Pending
902	Pending	Pending	Pending
903	Pending	Pending	Pending
904	Delete Transaction - Approved by Checker (Final Status)	Delete Transaction - Approved by Checker (Final Status)	Delete Transaction - Approved by Checker (Final Status)
905	Pending	Pending	Pending
906	Pending	Pending	Pending

STATUS Code	B2C Transactions – A Type
40	Pending at B2C Life Cycle
41	Available for PSU Maker
42	Available for PSU Checker
43	Available for FIG Maker
44	Available for FIG Checker
45	Available for Compliance Maker
46	Available for Compliance Checker
47	Rejected by PSU Maker
48	Approved by PSU User
49	Referred to Both (FIG & Compliance)
50	Rejected by PSU User

4.2 Error and Rejected Status Code / Return Message

STATUS CODE	DESCRIPTION	Account Validation	Balance Inquiry	Post Transaction	Get Status
1001	Transaction Reference number cannot be null	X	X	√	X
1002	Remittance Amount cannot be less than Or Equal to Zero (0)	X	X	√	X
1004	Invalid Remittance Type	X	X	√	X

1006	Remittance Amount cannot be less than Or Equal to Zero (0)	X	X	√	X
1007	Sender Name cannot be null	X	X	√	X
1008	Remittance Date cannot be null	X	X	√	X
1009	Beneficiary name cannot be null	X	X	√	X
1011	Invalid UBL Bank Code (D Type) OR Invalid Other Bank Code (C Type/Account Validation)	√	X	√	X
1013	Account Number cannot be null	X	X	√	X
1014	Branch Code contains characters.	X	X	√	X
1015	Account Number contains characters	X	X	√	X
1016	Invalid Branch Code.	√	X	X	X
1017	Invalid Account Number	√	X	√	X
1019	Duplicate Transaction Reference Number not allowed.	X	X	√	X
1020	Provided Information is Incorrect.	√	√	√	√
1027	Other Bank Information is incomplete	X	X	√	X
1030	IBFT Transactions Not Allowed For this Remitter.	X	X	√	X
1031	IBFT Account Format not Matched	X	X	√	X
1032	Invalid data length	X	X	X	X
1033	Transaction Not Found.	X	X	X	√
1035	Beneficiary account is commercial	X	X	√	X
1037	Invalid NON-PRI Bank	X	X	√	X
420	Invalid Sender Country	X	X	√	√
1018	Invalid Currency Code	X	X	√	X
1029	Transaction Amount is greater than the Limit (Only for Cash Type Transactions)	X	X	√	X

1040	Only UBL transactions are Allowed in B2C. Please send correct bank code.	X	X	√	X
1041	B2C transactions are not Allowed for this remitter.	X	X	√	X
1042	Invalid B2C Tran Type Given.	X	X	√	X
1043	System Error (see details below)	X	X	√	X
98	Current Transaction Amount Breaching Allowed Limit	X	X	√	X
99	Beneficiary Allowed Limit Already Consumed	X	X	√	X
2000	System Error (see details below)	√	√	√	√

2000 - System Error

This error relates to a general exception in process of a request in SOA HRC API. It can occur due to various reasons, most common of which are mentioned here below:

- Backend database down.
- Transaction processing host system down.
- Failure in any of the backend resources.
- Data Not found

In case of occurrence of *2000-System Error* in any critical request; kindly contact support team with the RQUID returned in the response envelope.

4.3 Bank Code and Account Formats

Bank Code	Bank Name	IBFT Enabled	PRISM Enabled	Valid Account Length(s)
1	Allied Bank	√	√	14, 20
2	Askari Comm Bank	√	√	13, 14
3	Bank Alfalah	√	√	12, 14, 18
4	Bank Al-Habib	√	√	17
7	Faysal Bank	√	√	9, 10, 11, 12, 13, 14, 16
8	First Woman Bank	√	√	
9	Habib Bank/KONNECT	√	√	14,17
10	Industrial Dev. Bank of Pakistan	X	√	
12	Meezan Bank	√	√	14,15
14	MCB Bank Ltd	√	√	16
15	National Bank	√	√	
19	Soneri Bank	√	√	15
20	Bank of Khyber	√	√	
21	Bank of Punjab	√	√	17
24	United Bank Limited	√ (not applicable to C type IBFT)	√	12
25	Zari Tarkeyati Bank	√	√	
27	Al Baraka Bank	√	√	13, 23, 24, 30
29	Citi Bank	√	√	10
31	Deutsche Bank A.G	X	√	

36	Standard Chartered Bank	√	√	11, 24
40	Bank Islamic Pakistan	√	√	15
44	Jahangir Siddiqui (JS) Bank Limited	√	√	6
45	Dubai Islamic Bank	√	√	10
46	S.M.E. Bank Limited	√	√	
47	Habib Metropolitan Bank	√	√	19, 20
54	Silk Bank	√	√	14
56	Summit bank limited	√	√	20
57	Samba Bank limited	√	√	
58	Askari Islamic Bank	X	√	
60	Sindh Bank	√	√	
61	Telenor Micro Finance Bank/EASY PAISA	√	√	16, 15
62	Khushhali Bank Limited	√	√	
63	First Micro Finance Bank	√	√	
64	Apna Micro Finance Bank	√	√	
65	MCB Islamic Bank Limited	√	√	
66	ICBC Bank Pakistan	√	√	
67	Finca Micro Finance Bank	√	√	
68	NRSP Microfinance Bank	√	√	
69	Advans Pakistan Microfinance Bank	√	√	
70	U Microfinance Bank Limited	√	√	
71	MOBILINK MICROFINANCE BANK/JAZZ CASH	√	√	
72	NAYAPAY BANK	√	X	
73	FINJA	√	X	
74	PAYMAX	√	X	
75	SADAPAY	√	X	

4.4 Authentication (OAuth) Steps

Before calling any service URL, the client must first get OAuth token form the URL:

<https://dpgwtdl.ubl.com.pk /tdl/sandbox/oauth2/token>

Request Header:

Attribute	Type	Cardinality	Description
Content-Type	String	1	"application/x-www-form-urlencoded"

Request (x-www-form-urlencoded):

Attribute	Type	Cardinality	Description
grant_type	String	1	"client_credentials"
client_id	String	1	Provided to you by UBL
client_secret	String	1	Provided to you by UBL
scope	String	1	Provided to you by UBL

Attribute	Type	Cardinality	Description
token_type	string	1	"Bearer"
access_token	string	1	String token that will be passed in the next URL
expires_in	number	1	Alive time for the token
consented_on	number	1	Time of consent
Scope	string	1	Scope that was provided in the request

After receiving the token from the above mentioned service, pass it in the request header for the next URL to be called. Token is added in "Authorization" header with prefix "Bearer[space]". Also, pass the client id in your next URL header with key value "X-IBM-Client-Id".

Example Request Header:

Attribute	Type	Cardinality	Example
Authorization	string	1	Bearer 1234567890qwertyuiopasdfghjkl;zxcvbnm, asdjklwekrjwiljcwejrlij3lj453.j4kl3j45
X-IBM-Client-Id	string	1	Jk12h31k-k1j2-v4u5-dsf9-4n3bk345nm0e